SPACE Y: Mars Habitat Control System

Decoupled IoT Platform for Environmental Monitoring & Automation

GROUP MEMBERS:

- Flavio Canovi - Automation Engine & Database Management
- Lorenzo D'Antonio - Data Ingestion & Streaming Services
- Christian Totaro - Frontend Dashboard & Architecture

THE CHALLENGE AND OUR APPROACH

The GOAL:

- Design a highly reactive and fault-tolerant control system for a simulated Martian habitat.
- Ensure seamless integration of heterogeneous data streams (REST & SSE).

Our SOLUTION:

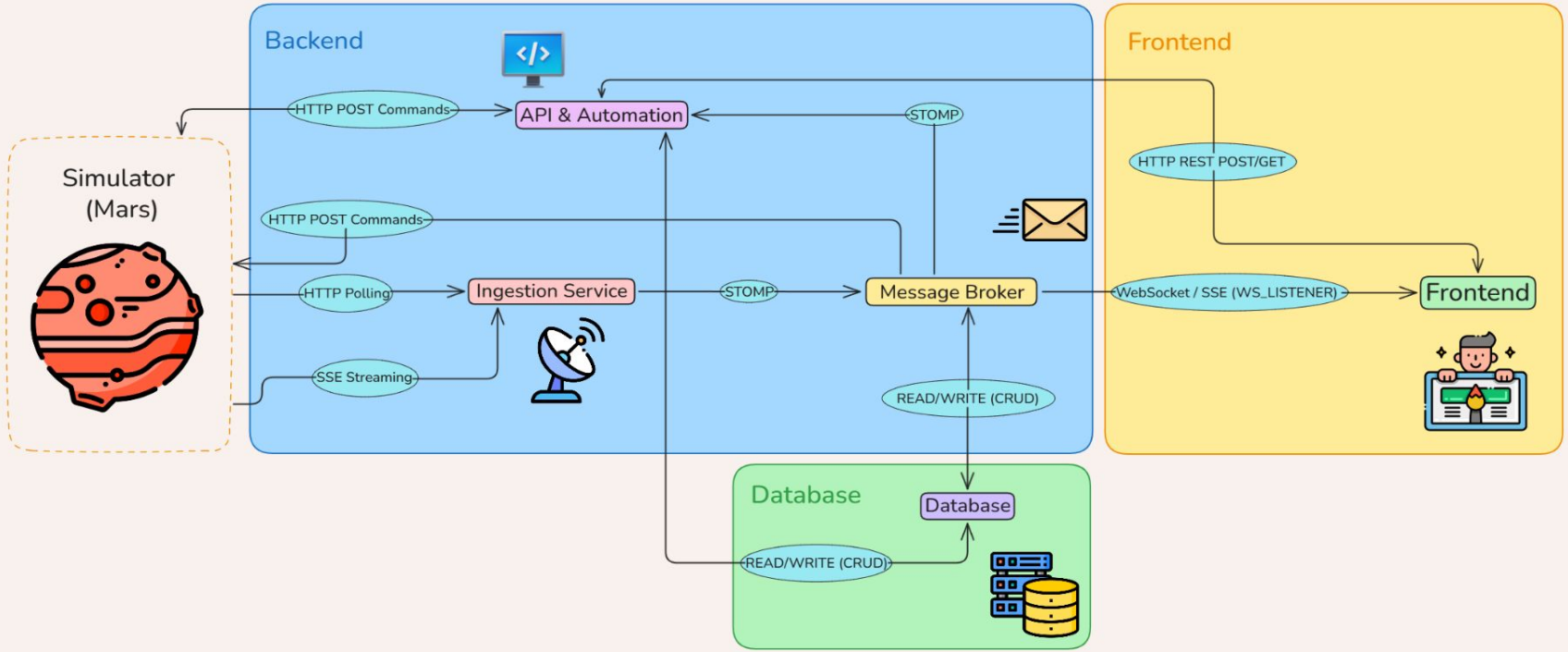
- Decoupled Microservices: The system is split into specialized, independent Docker containers.
- Asynchronous Messaging: Apache ActiveMQ handles real-time telemetry via STOMP.
- RESTful APIs: Used strictly for state configuration, rule management, and actuation.

System Architecture

Data Flow Overview:

- Ingestion: Dedicated services fetch data via REST Polling and SSE.
- Routing: ActiveMQ acts as the backbone, decoupling producers from consumers.
- Automation: FastAPI Engine consumes telemetry and evaluates SQLite rules.
- UI: NiceGUI Frontend subscribes directly to the Broker for real-time telemetry.

Architecture Diagram



Standard Internal Event Format

StandardFormat:

- id: str
- metric: str
- timestamp: datetime
- value: float
- unit: Optional[str] = None
- origin: str
- status: Optional[str] = None

User Stories

1. As a user, I want to view all the sensors and their live telemetry in a compact dashboard
2. As a user, I want to see clear visual status indicators for each sensor
3. As a user, I want to read the real-time values and unit of measurements for every metric
4. As a user, I want to view historical data charts for key metrics
5. As a user, I want to view the current state (ON/OFF) of every actuator
6. As a user, I want to manually toggle the state of the actuators
7. As a user, I want to be visually notified when an actuator changes state due to a rule
8. As a user, I want to see a dedicated page with a list of all active automation rules
9. As a user, I want to create a new rule
10. As a user, I want to modify the parameters of an existing rule
11. As a user, I want to delete a rule
12. As a user, I want all the rules to be saved automatically in a persistent database
13. As a user, I want the system to automatically block the creation of a new rule if its logic overlaps with an existing one
14. As a user, I want the system to automatically block the modification of a rule if the updated parameters conflict with another existing rule
15. As a user, I want to be visually notified when I create, modify or delete a rule

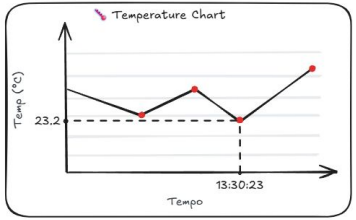
Dashboard

Environmental Sensors

Greenhouse Temp
25.5 C°
OK

Entrance Humidity
30 %
OK

Data Graphs



Actuators Control

Ventilation
ON

Heater
OFF

On Rule triggered

Automation: Hall Ventilation -> ON

Press

Manual: Heater -> ON

Live Telemetry

Solar Panels
Power: 15.5 kW
Voltage: 120.0 V
Current: 129.1 A
Total: 3450 kWh
OK

Thermal Loop
Temp: 45.2 °C
Flow: 15.0 L/min
OK

Radiation
Level: 0.08 uSv/h
OK

Technical Highlights & Solutions

- Async Multi-Stream Management: Implemented "asyncio" to simultaneously handle REST polling and SSE streams without blocking.
- Algorithmic Conflict Prevention: A custom module calculates overlaps, blocking contradictory rules before database insertion.
- Reactive "Dumb" Frontend: The UI updates instantly via STOMP/WebSockets but contains zero business logic (Single Source of Truth).

System in Action: Live Demo